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Eagle 250
Installation and Technical Manual

DSM-00222-01

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For AM-3500-E100, -E200, -E300, -E400, -E500 and AM-990-01 systems replace battery with Panasonic or Ray-O-Vac BR2325 only. For AM-3500-250, AM-3500-E450, AM-3500-E550, AM-3500-6000, and AM-990-04 systems, replace batteries with Panasonic or Ray-O-Vac BR1225 only. Use of other batteries may present a risk of fire or explosion. Replacement batteries may be ordered from your authorized Alpha Micro reseller.

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This computer contains no user-configurable components that require opening the computer case. Because the power supply in this computer is capable of outputting high current levels hazardous to your safety, the computer case should only be opened by an authorized service technician.

Cet ordinateur ne contient aucune pièce configurable par l'utilisateur qui nécessite l'ouverture du boitier. L'alimentation de cet ordinateur peut preduire des nivaeux de tensions dangereux, le boitier ne devrait donc être ouvert que par un technician autoriaé.

SOFTWARE SECURITY DEVICE IDENTIFICATION NUMBER: _____

The Alpha Micro Software Security Device (SSD) is a customized integrated circuit that personalizes the computer, providing identity verification for it. Certain Alpha Micro and non-Alpha Micro software may require that your computer contain an SSD in order to run software that has been customized to run only on your computer.

Please enter the identification of your SSD above. The SSD identification number should be on your computer ID label under "SSD Serial No." (Another way of finding the number is to look at the SSD itself. The SSD is located in an integrated circuit location on the CPU board; its identification number is printed on the SSD itself.) Software vendors may ask you for the SSD number if they are customizing software to run only on your computer.

This document may contain references to products covered under the following U.S. Patent Number(s): 4,530,048

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Chapter 1 - Introduction

This manual is for Alpha Microsystems dealers and technical personnel: people who will sell, install, and maintain Eagle 250 computers. It does not contain material which the average end-user needs to know, and, in general, does not need to be given to the end-user.

This book makes some assumptions about you, the reader:

- You are familiar with today's computer terminology: you know what terms such as serial I/O, Ethernet, and disk cache mean.
- You have used AMOS before, and have at least some knowledge of the operating system and of previous Alpha Micro products. For example, you do not need to be told what the system initialization command file is.

In addition to this introduction, which describes this book and the other Eagle 250 documentation, this manual includes:

- **Chapter 2 - Specifications**: information which was contained in a separate specifications document for previous systems: size and weight, electrical requirements, processor capabilities, and so on.
- **Chapter 3 - Configuration**: guidelines for designing an Eagle 250: what I/O boards you can use, how many SCSI devices you can hook up, etc.
- **Chapter 4 - Installatio**n: site requirements, CMOS and initialization file configuration, and other installation procedures.
- **Chapter 5 - Troubleshooting**: status display codes and troubleshooting information.
- **Appendix A - Read-ahead and Write Buffering**: what they are and how to use them to optimize system performance.
- **Appendix B - Super I/O**: how to turn off this performance-enhancing feature if necessary.

THE EAGLE 250 DOCUMENTATION SET

The complete documentation package for the Eagle 250 computer consists of three manuals, each containing a different type of information and aimed at a different audience:

- *The Eagle 250 Owner's Manual*: for the end-user, it contains non-technical information about getting started with AMOS, how to care for the computer, and what to do if something goes wrong. We recommend that you give this manual to the end user; it will answer many of their questions, and it does not include technical information.
- *The Eagle 250 Installation and Technical Manual* (this book): for the dealer, with technical specifications and configuration information and installation instructions. We do not suggest you forward this book to the customer.
- *The Eagle 250 Service Manual*: for qualified service personnel only: opening the chassis, installing new devices and changing board jumper settings.

Graphic Conventions

Like other Alpha Micro documents, this manual uses some standard symbols and special typefaces to make our examples and explanations easier to read and understand:

Symbol	Description
	This means **STOP!**, and signals an important warning or restriction. Be sure to read the text next to this symbol carefully, as it could help you avoid serious problems.
	This marks a hint—a shortcut or an easier way to do something.
	This indicates a note: information which relates to the current topic, and may be important for you to remember.
`Text`	We show characters the computer displays on your screen, such as prompts and information messages, in this typeface. We also use it for command file examples.
`TEXT`	In examples, we use this typeface for the characters you type on your keyboard.
KEY	This symbol tells you to press the indicated key. For example: **`DIR`** ENTER tells you to press the ENTER key at the end of the DIR command.
CTRL/C	This combination of symbols tells you to hold down the first key and press the second key. For example, to type a CTRL/C (Control-C), press the CTRL key and, while holding it down, type a **`C`**.

Chapter 2 - Specifications

GENERAL SPECIFICATIONS

Temperature/Humidity	Temperature: 60° to 90° F (15.6° to 32.2° C); 10° C per hour maximum fluctuation; Humidity: 20% to 80%, non-condensing
Power Requirements	USA and Canada: 115 VAC (90 to 132 VAC), 47 to 63Hz International: 230 VAC (180 to 264 VAC), 47 to 63Hz
Maximum AC Power Consumption	AM-138 board: 5 amps at 115 volts or 2.5 amps at 230 volts Note: due to varying peripherals, it is impossible to determine total power requirements for complete system.
Maximum DC Power Requirements	+5 volts: 4.3A with 256MB DRAM; 3.9A with 8MB DRAM +12 volts: less than 100mA Figures for AM-138 board only
Electromagnetic Interference	Complies with US FCC rules and regulations, Class A
System Dimensions	Height: 17.75” (44.5 cm) Width: 9” (22.9 cm) Depth: 17” (43.1 cm) Weight: Up to 26 lb. (12 kg)
Enclosure Composition	Sheet metal and plastic
Additional Features	Clock/calendar with battery backup Front panel with status and diagnostic display 32-bit, bus-master RISC SCSI controller Programmable interval timer AC power failure detect Remote reset option Security lock CMOS setup feature Dedicated UPS monitoring port Optional AM-338 RJE interface

REAR PANEL CONNECTIONS

Serial Ports (Full Eagle 250)	Eight standard, with full modem control. DB-9 connector configuration Up to 32 possible using AM-318-10 serial I/O boards and AM-90 lightning protection boards (see below) Also accepts AM-314 and AM-318-00 boards.
Serial Ports (Eagle 250LC)	Four standard, with full modem control. DB-9 connector configuration. Optional AM-90 lightning protection board available. No serial I/O expansion for this model.
Parallel Port	One 25-pin (see below)
Ethernet Port	10BaseT
External SCSI Port	Narrow or Wide, for external bus termination or external devices
UPS Monitoring Port	9-pin

AM-138 CENTRAL PROCESSING UNIT

Processor	Motorola MCF5102 Coldfire CPU with variable-length RISC
CPU Speed	33MHz
Data/Address Path	Multiplexed 32-bit data/address path
Internal Cache	2KB instruction cache
Memory Expansion	Two SIMM slots; from 4MB to 256MB; can install either one or two SIMMs
Interrupt Capabilities	Seven interrupt levels with vector capability
Direct Memory Access	Bus mastering capability, supports multiple masters
Ethernet	32-bit dual-port controller
SCSI Controller	32-bit bus-master RISC processor

STORAGE AND BACKUP

Fast-Wide SCSI (Small Computer System Interface) Bus

Physical interface	68-pin connector
Maximum Bus Length	Ten feet (three meters)
Maximum Number of Devices	Up to five internal; multiple external devices (eight feet of cable) possible using AM-441 Wide SCSI Bus Repeater
Protocol Supported	Full SCSI-1, SCSI-2 and Fast/Wide-SCSI supported, including support for connect and disconnect protocols
Implementation	Single host only; multiple host architecture not supported

Narrow SCSI Bus

Physical Interface	50-pin connector
Maximum Bus Length	20 feet (six meters)
Maximum Number of Devices	Eight unique SCSI addresses, including the host controller
Protocol Supported	Full SCSI-1 and SCSI-2 supported, including support for connect and disconnect protocols
Implementation	Single host only; multiple host architecture not supported

The two busses (SCSI and Wide-SCSI) are mutually exclusive. All devices must connect to one of the two busses; the other connector cannot be used. Using the appropriate adapters, you can connect a mix of SCSI-2 drives and Wide SCSI-2 drives to either SCSI bus.

Hard Disk Drives

Alpha Micro offers a complete line of narrow and wide SCSI disk drives, including external RAID subsystems. See the current AMOS Hardware price list for the drives available.

Physical Size	3.5” or 5.25”
Maximum Number	Internal: five half-height; two can be either 3.5” or 5.25”; others must be 3.5”; more may be added externally, depending on bus limitations
Interface	SCSI or Wide SCSI

Streaming Tape Drives

Physical Size	5.25” half-high
Interface	SCSI or Wide SCSI
Data Capacity	150MB to 26GB
Tape Cartridge	See *Eagle 250 Computer Owner’s Manual*
Tape Format	See *Eagle 250 Computer Owner’s Manual*
Soft Error Rate	1×10^{-8}
Hard Error Rate	1×10^{-10}

Diskette Drive

Physical Size	3.5”, but requires 5.25” mounting bay
Interface	SCSI; AM-219 Floppy Controller is ***not*** used
Data Capacity	1.44MB or 720KB depending on the format.
Diskettes	3.5” 18 sectors/track, 80 tracks/side, double-sided, double-density
Transfer Rate	500K bits/second.
Latency (Average)	100 msec.
Access Time	Track-to-track: 3 msec. Average: 94 msec.
Track Density	135 TPI.
Recording Density	17,434 bits/inch.
MTBF	10,000 power on hours or more, normal duty cycle.
Error Rates	Soft read: 1×10^{-9} Hard read: 1×10^{-12} Seek: 1×10^{-4}

SERIAL I/O

Standard (Full Eagle 250)	Eight ports, with flow control, DB-9 connectors.
Standard (Eagle 250LC)	Four ports, with flow control, DB-9 connectors
Characteristics	All asynchronous serial communication ports configured for RS232. RS422 is not supported.
Interface	DB-9 connectors standard. 8-wire RJ-45 jacks optional. For pin signals see Chapter 3.
Supported Baud Rates	50, 75, 110, 134.5, 150, 300, 600, 1200, 1800, 2000, 2400, 3600, 4800, 7200, 9600, 19200, 38400, 57600
Stop Bits	2 stop bits if 110 baud; 1 stop bit all others—default.
Word Length	Default: 8 data bits
Parity	Default: No parity
Expansion (Full Eagle 250 only)	Three expansion slots for up to 24 ports total, using AM-318-10 boards, or 32 ports if AM-90 Lightning protection board option is used with the first 8 ports.

AM-318-10 Serial I/O Board (Full Eagle 250 Only)

Number of ports	Eight per board; total of up to 24, with flow control and lightning protection
Characteristics	All asynchronous serial communication ports configured for RS232. RS422 is not supported.
Interface	8-wire RJ-45 jacks standard. For pinout diagram, see Chapter 3.
Supported Baud Rates	50, 75, 110, 134.5, 150, 300, 600, 1200, 1800, 2000, 2400, 3600, 4800, 7200, 9600, 19200, 38400, 57600
Stop Bits	2 stop bits if 110 baud; 1 stop bit all others—default
Word Length	Default: 8 data bits
Parity	Default: No parity

For compatibility with earlier systems, the Eagle 250 also supports 4-port AM-314 and 8-port AM-318-00 serial I/O boards, with or without AM-90 cards for lightning protection.

PARALLEL I/O

Number of Ports	One
Characteristics	Centronics compatible
Interface	DB-25 Connector; for DB-25 connector signals, see Chapter 3
Cable length	Parallel printer cable should not exceed six feet

Chapter 3 - Configuration

This chapter discusses configuration rules, considerations, and procedures for the Eagle 250. It covers:

- Back panel connectors
- Wide and narrow SCSI configuration
- Serial and parallel I/O, including cabling information
- Networking
- UPS status port

THE EAGLE 250 BACK PANEL

The illustration on the next page shows the Eagle 250 back panel, with the usable connectors labeled. Refer to that drawing for the location of these connections:

- The standard four or eight serial ports are shown in DB-9 connector configuration. To achieve maximum expansion potential for the full Eagle 250, optional AM-90 Lightning protection boards can be installed in place of the DB-9 connectors to provide eight RJ-45 connectors in I/O slots 0 and 1 on the rear panel, leaving I/O slots 2 through 7 for serial I/O expansion as described below. The Eagle 250LC is not expandable beyond the standard four serial ports.
- Each serial I/O expansion slot will accept an AM-90 board, which provides four RJ-45 connectors (standard for the eight on-board serial ports and the AM-318-10), an RJ-21 Telco connector from an AM-318-00, or, with an adapter, two DB-9 connectors from an AM-314. If you choose to use DB-9 connectors for compatibility with earlier hardware, your total port capacity is reduced, as only two ports fit in each slot instead of four.
- The Ethernet port supports 10BaseT connection only.
- The external SCSI connector may be either a Wide SCSI-2 68-pin or a SCSI-2 50-pin connector, depending on the SCSI bus you've chosen.

These notes refer only to the types of connections you may see on the back panel; for more information on SCSI, serial, and parallel I/O, and network connections, please refer to the appropriate sections later in this chapter.

This back panel is used for factory-shipped Eagle 250s. Systems field-upgraded to an Eagle 250 have a different back panel, as shown in PDI-00138-15, *Upgrade Instructions, Eagle 100 to Eagle 250*.

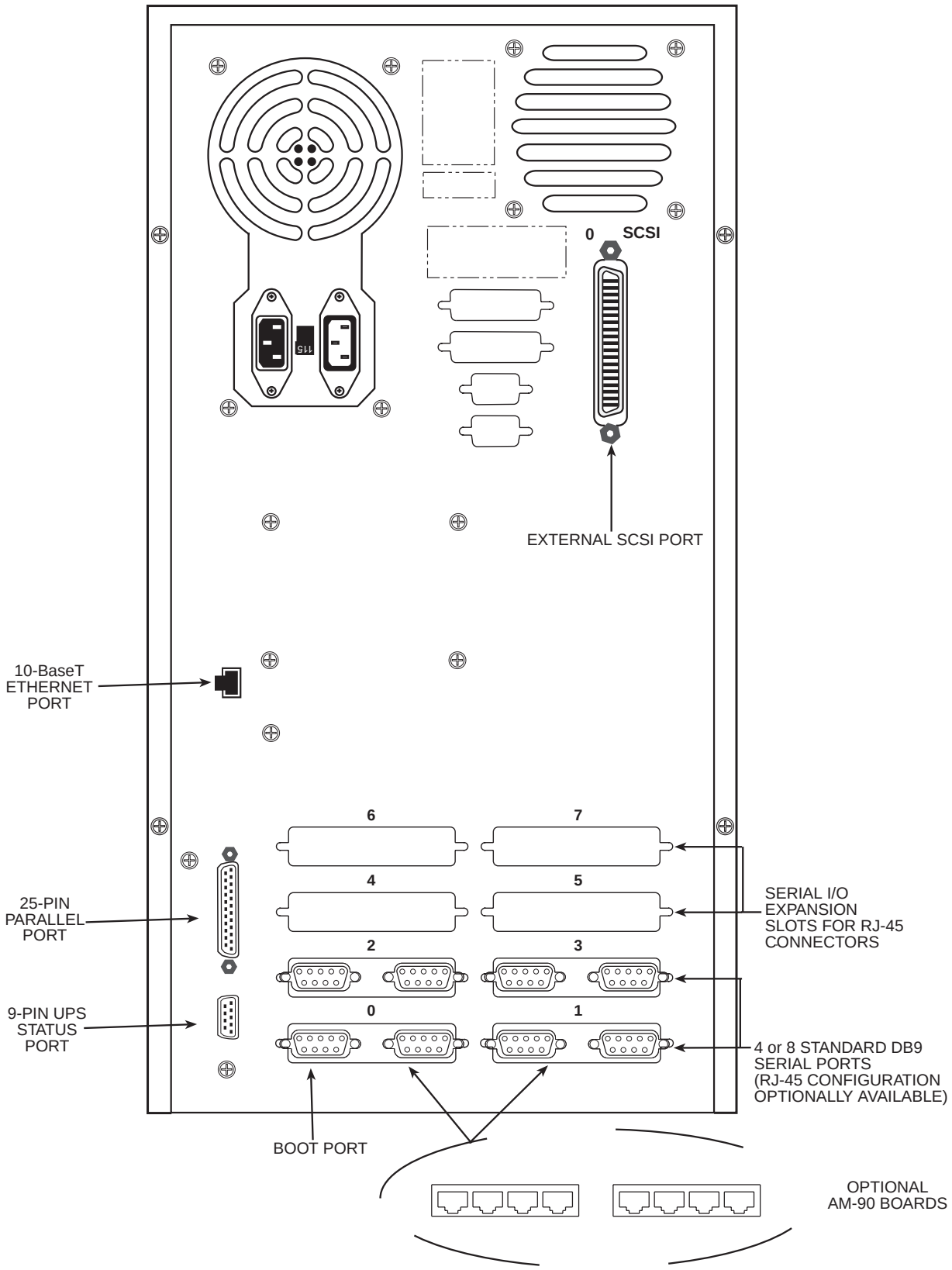

Figure 3-1: Eagle 250 Back Panel

SCSI CONFIGURATION

The Eagle 250 offers two SCSI busses: Wide and narrow. The Wide SCSI bus offers higher performance when used with Wide SCSI-2 or Ultra SCSI drives. Using the proper adapters, you can attach both Wide and narrow devices—disk drives, tape drives, and SCSI diskette drives—to either the Wide or narrow bus. However, using narrow devices on the Wide bus may affect the performance of Wide drives on the bus.

You can use only one of the two SCSI busses. You cannot attach cables to both connectors and use both busses at once.

The number of devices you can attach to the SCSI bus can be limited by three factors: the legal number of SCSI IDs, cabling limitations—total length and device spacing—and the number of drive bays available in the chassis. After a discussion of the type of SCSI devices you can use, the following sections describe configuration rules for the Wide and narrow busses, including mixing device types and termination issues.

Never plug a SCSI device into the SCSI cable, or remove one from the cable, while system power is on. Doing so could seriously damage the device and/or the CPU board. The Eagle 250 uses tolerant active negation on the SCSI bus, which makes the bus more sensitive in these situations than on previous Alpha Micro computers.

SCSI Device Types

You can attach any supported SCSI device to either the Wide or narrow SCSI bus, using the appropriate adapter if necessary, as described below. Supported devices include:

- Narrow, Wide and UltraSCSI disk drives
- ¼” streaming tape drives with capacities up to 26GB
- 4mm DAT (Digital Audio Tape) drives
- The AM-446 RAID subsystem for the safety of redundant data storage
- The AM-642 StoP (SCSI to Pertec) converter to use ½” magnetic tape as a SCSI device
- CD-ROM drives
- 3.5” SCSI diskette drives. The Eagle 250 is the first AMOS system to support SCSI diskette drives, removing the need for a separate diskette controller.

Alpha Micro offers a wide array of both narrow and Wide SCSI disk drives and tape drives. For current product offerings and prices, please see the latest AMOS Hardware Price List.

Wide SCSI Configuration Rules

The Wide SCSI bus allows up to 15 SCSI IDs, so the limiting factors are the number of device bays in the chassis and, especially, cabling considerations. ***Because of the higher performance of the Wide bus, it is critical to follow the cabling specification strictly, or performance and reliability will suffer.***

The two cabling specifications which affect the total number of devices are:

- The total length of the cable cannot exceed three meters (approximately 10 feet).
- Each device must be at least one foot (approximately) from any other device.

The standard internal Wide SCSI cable for the Eagle 250 ensures adequate space between internal devices. It allows up to five internal SCSI devices, plus the external connector. While this connector is normally used for an active terminator, you can attach an external cable to another SCSI device. To remain within the specification, the maximum length of the external cable is three feet. In practice, this normally means only one external device is possible. If you need a longer external cable, use a repeater, as described below.

To summarize, unless you use a repeater, the Eagle 250's Wide SCSI bus supports up to five internal devices and one external device.

If you are attaching an external device, especially if it is a narrow device, be sure to read the section on Wide Bus Termination, below.

The Wide SCSI Repeater

As mentioned above, the total allowable bus length for the Wide SCSI bus is 10 feet. This can be very limiting, especially in configurations which require more than one external device. The AM-441 Wide SCSI Bus Repeater attaches to the end of the Eagle 250's internal SCSI cable. In effect, it starts a new physical bus: from the repeater, you can have up to ten additional feet of bus cable.

Allowing for the internal cable from the repeater to the external SCSI port, the AM-441 lets you have up to eight feet of external Wide SCSI cabling, with as many devices as you can physically attach, obeying the specification of at least one foot between each two devices.

The AM-441 occupies an internal 5.25" drive bay. Since there are six available bays (three of them 5.25") and only five available connectors on the internal SCSI cable, this does not reduce the number of internal SCSI devices you can use (the AM-441 attaches to the connector which is normally used for the external port, so it does not take up an internal drive connector).

Narrow Devices on the Wide Bus

You can attach any narrow SCSI device to the Wide bus by using a 50-pin to 68-pin adapter, PDB-00440-91, between the device and the cable connector. However, if you use both a narrow and a Wide disk drive, the narrow drive's lower transfer rate slows down the entire bus, causing the Wide drive to lose its performance advantage. For best results, we recommend using only Wide disk drives on the Wide bus.

On the internal cable, the order of Wide and narrow devices does not matter. If you use both Wide and narrow external devices, the narrow device(s) must be the last device(s) on the external cable. See the next section on how to properly terminate the bus if you have narrow external devices.

Wide Bus Termination

When terminating the Wide SCSI bus, keep two things in mind:

- All 16 lines of the bus must be terminated.
- Termination must be active.

If you have only internal SCSI devices, termination is simple: just make sure the external active terminator supplied with the Eagle 250 is installed properly. Similarly, if you have only Wide external devices, simply remove the terminator from the external port on the chassis and place it in the unused connector on the last device on the external cable.

When you have a narrow external SCSI device, such as a CD-ROM drive, termination becomes slightly more complicated. As stated above, any narrow external device must be the last device on the cable, beyond any Wide external devices. You must terminate the “high” nine lines of the Wide bus before the first external narrow device, and the rest of the bus signals at the last device. To do this:

1. Between the last Wide device and the first narrow device (if you have only narrow external devices, between the external Wide connector and the first narrow device), you must use a Wide-to-narrow cable which actively terminates the high nine lines. Alpha Micro offers two such cables: PDB-00440-80 (3 ft.) and PDB-00440-81 (6 ft.). You cannot use the six-foot cable unless you are using the AM-441Wide SCSI Bus Repeater.
2. Plug an active narrow terminator into the unused SCSI connector on the last narrow device. One is available from Alpha Micro, part number PRA-00222-21.

In this configuration, the external Wide terminator included with the Eagle 250 is not used.

Narrow SCSI Configuration Rules

The narrow SCSI bus allows a cable length of up to 6 meters (approximately 20 feet). This means that, unlike the Wide bus, total cable length is rarely a consideration. Instead, the limiting factors are the number of connectors on the internal cable and the allowed number of device IDs.

The narrow bus only allows IDs from 0 through 7. Since ID 7 is used by the host controller, ***you cannot have more than seven devices on the narrow bus***.

The internal SCSI cable shipped with your Eagle 250 has only four drive connectors, so you cannot have more than four internal devices. You can, however, easily have four internal devices and three external ones. The external devices can be “stand-alone” devices like the AM-642 SCSI to Pertec converter, or they can be in an expansion chassis such as the AM-3501.

Wide Devices on the Narrow Bus

You can attach any Wide SCSI device—disk drive or tape drive— to the narrow bus by using a 68-pin to 50-pin adapter, PDB-00440-90, between the device and the cable connector. However, on the narrow bus, Wide devices do not perform any better than comparable narrow devices. By using the narrow bus you sacrifice any performance advantage.

Narrow SCSI Termination

In most cases, terminating the narrow SCSI bus is simple: plug the active narrow terminator (PRA-00222-21) supplied with your computer into the external SCSI connector or, if you have one or more external devices, into the unused SCSI connector on the last device.

The only exception to this is if you have an external Wide SCSI device as the last device on the narrow cable. In this case, you must get an active Wide SCSI terminator, part number PRA-00222-20, and plug it into the unused SCSI connector on the last Wide device.

SERIAL I/O CONFIGURATION

The AM-138 CPU board in the Eagle 250 includes eight on-board serial ports, plus three I/O expansion slots. The on-board ports support full modem control and have DB-9 connectors. Optional AM-90 Lightning cards are available to provide surge protection and RJ-45 jacks for these ports on the back panel. The AM-138 board in the Eagle 250LC includes four on-board board serial ports, and does not include any expansion slots

The following sections discuss the boards and board combinations supported in the full Eagle 250 expansion slots, signal pinouts, and serial cable packages available from Alpha Micro.

You can find information on software setup of the serial ports, including the interface driver names to use, in Chapter 4.

Serial I/O Boards Supported

You can use any of these serial I/O cards in the three expansions slots of the Eagle 250:

- **AM-318-10**: This 8-port board is the preferred I/O board for the Eagle 250. It supports modem control and has built-in lightning protection from the included AM-90 Lightning cards. It occupies two back panel slots, each with four RJ-45 jacks.
- **AM-314**: This older 4-port board also supports modem control. In its standard configuration, it uses a DB-9 connector for each port, occupying two back panel slots for each 4-port board. You can attach an AM-90 to it, adding lightning protection and changing the back panel configuration to one slot with four RJ-45 jacks. The AM-314 does not support the Super I/O software.
- **AM-318-00**: An 8-port board without modem control or lightning protection. All eight ports are contained in a single RJ-21 back panel connector.
- **AM-318-02**: An AM-318-00 with two AM-90 cards added, providing lightning protection and changing the back panel configuration to eight RJ-45 jacks in two slots.

While you can use these boards in any combination, the AM-318-10 is preferred because it includes the most features—full modem control, lightning protection, and Super I/O support—and the highest number of ports—eight per expansion slot. The other AM-318-xx boards do not offer modem control; while the AM-314 provides only four ports per expansion slot and doesn’t support Super I/O.

If you do mix board types, you can put any supported I/O board in any expansion slot. However, be careful to use the correct port numbers when defining the ports in the initialization file, as discussed in

mahChapter 4. We do recommend placing any AM-314 boards after any other board(s) in the computer, but this is not required.

If you are going to use the older AM-314 or AM-318-00 boards in an Eagle 250, we recommend adding AM-90 Lightning boards to them and to your on-board 8 ports, especially if you live in an area prone to thunderstorms. Even if lightning isn't a frequent problem for you, the AM-90 may simplify cabling, and it never hurts to have extra protection against power surges.

You can also install an AM-338 RJE card into one of the serial I/O expansion slots. This board provides a bisync interface for RJE communication with mainframe computers. Since it occupies an expansion slot, it reduces the maximum number of serial ports you can install to 24.

Signal Pinouts

All serial ports on the Eagle 250 support only RS-232 communication; they do not support RS-422.

The basic on-board serial I/O ports and the AM-314 use this pinout scheme for their DB-9 connectors:

Pin #	Description	
1	Unused	
2	Receive data	RXD
3	Transmit data	TXD
4	Clear to send	CTS
5	Request to send	RTS
6	Unused	
7	Signal ground	GND
8	Data carrier detect	DCD
9	Data terminal ready	DTR

Serial I/O ports on the AM-318-10 and AM-90 Lightning protection boards use the same pinouts for their RJ-45 jacks (these pinouts are correct for any AM-90 RJ-45 port; they are the same as for the AM-359 card):

Pin #	Description	
1	Chassis ground (shield)	GND
2	Clear to send	CTS
3	Transmit data	TXD
4	Request to send	RTS
5	Receive data	RXD
6	Data terminal ready	DTR
7	Signal ground	GND
8	Data carrier detect	DCD

Pins are numbered from left to right, looking into the port from outside the computer chassis.

All 8 ports on the AM-318-00 are contained in one 50-pin RJ-21 connector, using these pinouts:

Port	Signal		Pin #	Port	Signal		Pin #
0	Receive data	RXD	26	**4**	Receive Data	RXD	38
	Request to send	RTS	1		Request to send	RTS	13
	Transmit data	TXD	27		Transmit data	TXD	39
		NC	2			NC	14
	Clear to send	CTS	28		Clear to send	CTS	40
	Signal ground	GND	3		Signal ground	GND	15
1	Receive Data	RXD	29	**5**	Receive Data	RXD	41
	Request to send	RTS	4		Request to send	RTS	16
	Transmit data	TXD	30		Transmit data	TXD	42
		NC	5			NC	17
	Clear to send	CTS	31		Clear to send	CTS	43
	Signal ground	GND	6		Signal ground	GND	18
2	Receive Data	RXD	32	**6**	Receive Data	RXD	44
	Request to send	RTS	7		Request to send	RTS	19
	Transmit data	TXD	33		Transmit data	TXD	45
		NC	8			NC	20
	Clear to send	CTS	34		Clear to send	CTS	46
	Signal ground	GND	9		Signal ground	GND	21
3	Receive Data	RXD	35	**7**	Receive Data	RXD	47
	Request to send	RTS	10		Request to send	RTS	22
	Transmit data	TXD	36		Transmit data	TXD	48
		NC	11			NC	23
	Clear to send	CTS	37		Clear to send	CTS	49
	Signal ground	GND	12		Signal ground	GND	24

For more information on cabling the AM-318-00, please see *Installation Instructions: AM-318 Serial I/O Controller*, PDI-00318-00.

Cabling Packages

The following cable packages are available from Alpha Micro to use with the RJ-45 jacks provided by the AM-90 boards included in your Eagle 250, the ports from an AM-318-10, or AM-314 or AM-318-00 ports using AM-90 cards. These are the same kits used for the AM-359 cards in large AMOS systems such as the AM-6000.

The cabling configurations described in this appendix meet the USOC (United Standards Organization Committee) Standards.

- **PDB-00359-50**—For configurations where the Eagle 250 ports are replacing an AM-355. The kit includes eight PRA-00243-04 cable assemblies. Each assembly has a four-foot parallel patch cord with an RJ-45 connector at one end and a modular female DB9 connector at the opposite end. You simply plug the RJ-45 connector into the Eagle 250 and the female DB9 into the male DB9 connector previously plugged into the AM-355 board.

- **PDB-00359-51**—Connects the Eagle 250 ports to an existing six-wire patch panel using a three-pair wiring system. The cable assembly has eight RJ-45 connectors with six wires each, which plug into the two AM-90 boards attached to Eagle 250 ports. The opposite end of the cable assembly is a female RJ-21 connector which plugs into your six-wire patch panel.

  This cabling kit (PDB-00359-51) does not support DCD for modems, but it does support DTR.

- **PDB-00359-52**—Connects the Eagle 250 ports to an eight-wire patch panel using a four-pair wiring system. The cable assembly has six RJ-45 connectors with eight wires each, which plug into six Eagle 250 RJ-45 connectors. The opposite end of the cable assembly is a female RJ-21 connector which plugs into your eight-wire patch panel. This kit also includes two of the four-foot long RJ-45 to female DB9 connectors used in the PDB-00359-50 cabling kit, for the two ports not plugged into the patch panel.
- **PDB-00359-53**—Four RJ-45 to male DB25 adapters for connecting modems to the Eagle 250. This kit requires a shielded parallel patch cord with RJ-45 connectors at each end, like the 10-foot PRA-00189-10 cable in the PDB-00359-56 cabling kit described below.
- **PDB-00359-54**—Four modular RJ-45 to male DB25 adapters for connecting printers and terminals to the Eagle 250 ports. This kit requires a shielded parallel patch cord with RJ-45 connectors at each end, like the 10-foot PRA-00189-10 cable supplied in the PDB-00359-56 cabling kit described below.
- **PDB-00359-55**—Four modular RJ-45 to female DB9 adapters for connecting a PC-AT style serial port to an Eagle 250 port. This kit requires a shielded parallel patch cord with RJ-45 connectors at each end, like the 10-foot PRA-00189-10 cable supplied in the PDB-00359-56 cabling kit described below.
- **PDB-00359-56**—This kit includes four (PRA-00189-10) 10 foot shielded parallel patch cords with RJ-45 connectors at each end.

The following illustrations contain signal and pinout information for each of the PDB-00359-5x cabling kits:

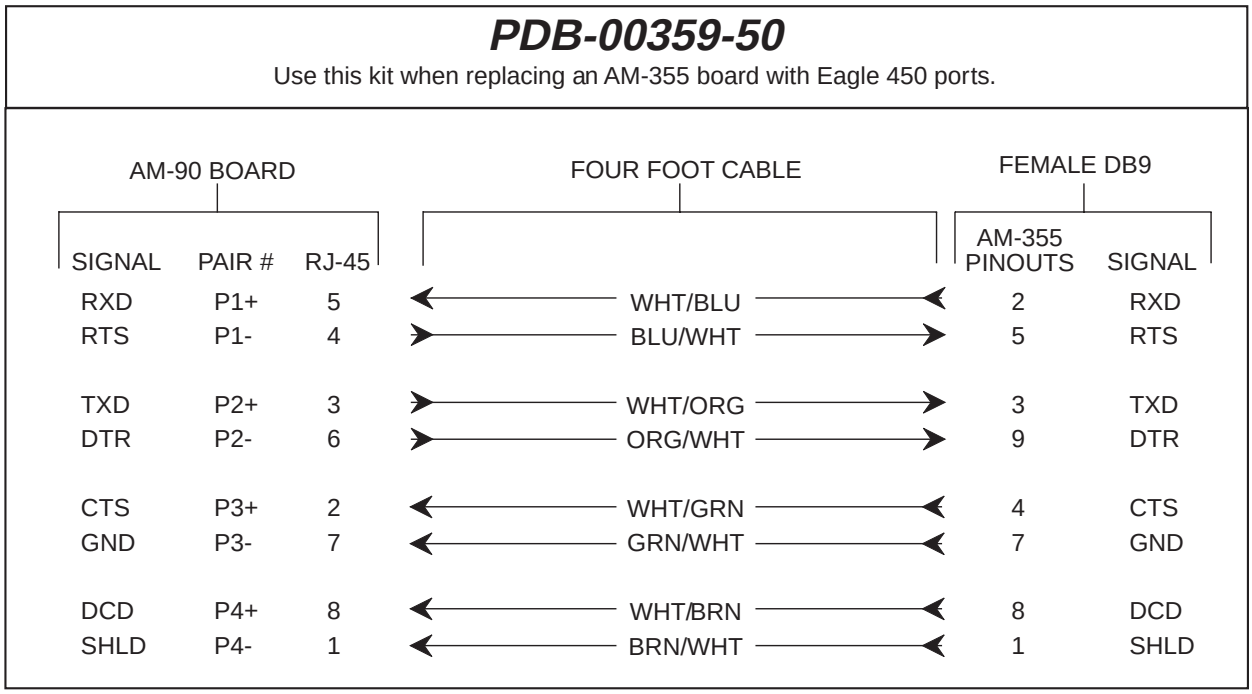

Figure 3-2: PDB-00359-50 External Cabling Kit

PDB-00359-51

Use this kit to attach an Eagle 450 to an existing six-wire patch panel using a three-pair wiring system.

	AM-90 BOARD PAIR #	RJ-45		FEMALE RJ21 CONNECTOR
PORT 1	P1+	5	WHT/BLU	26
	P1-	4	BLU/WHT	1
	P2+	3	WHT/ORG	27
	P2-	6	ORG/WHT	2
	P3+	2	WHT/GRN	28
	P3-	7	GRN/WHT	3
PORT 2	P1+	5	WHT/BLU	29
	P1-	4	BLU/WHT	4
	P2+	3	WHT/ORG	30
	P2-	6	ORG/WHT	5
	P3+	2	WHT/GRN	31
	P3-	7	GRN/WHT	6
PORT 3	P1+	5	WHT/BLU	32
	P1-	4	BLU/WHT	7
	P2+	3	WHT/ORG	33
	P2-	6	ORG/WHT	8
	P3+	2	WHT/GRN	34
	P3-	7	GRN/WHT	9
PORT 4	P1+	5	WHT/BLU	35
	P1-	4	BLU/WHT	10
	P2+	3	WHT/ORG	36
	P2-	6	ORG/WHT	11
	P3+	2	WHT/GRN	37
	P3-	7	GRN/WHT	12

	AM-90 BOARD PAIR #	RJ-45		FEMALE RJ21 CONNECTOR
PORT 5	P1+	5	WHT/BLU	38
	P1-	4	BLU/WHT	13
	P2+	3	WHT/ORG	39
	P2-	6	ORG/WHT	14
	P3+	2	WHT/GRN	40
	P3-	7	GRN/WHT	15
PORT 6	P1+	5	WHT/BLU	41
	P1-	4	BLU/WHT	16
	P2+	3	WHT/ORG	42
	P2-	6	ORG/WHT	17
	P3+	2	WHT/GRN	43
	P3-	7	GRN/WHT	18
PORT 7	P1+	5	WHT/BLU	44
	P1-	4	BLU/WHT	19
	P2+	3	WHT/ORG	45
	P2-	6	ORG/WHT	20
	P3+	2	WHT/GRN	46
	P3-	7	GRN/WHT	21
PORT 8	P1+	5	WHT/BLU	47
	P1-	4	BLU/WHT	22
	P2+	3	WHT/ORG	48
	P2-	6	ORG/WHT	23
	P3+	2	WHT/GRN	49
	P3-	7	GRN/WHT	24

Figure 3-3: PDB-00359-51 External Cabling Kit

PDB-00359-52

Use this kit to connect eight Eagle 450 ports to an eight-wire patch panel using a four-pair wiring system. This kit creates two extra eight-wire AM-355 type ports (ports 7 and 8), which have standard female DB9 connectors.

	AM-90 BOARD PAIR #	RJ-45		FEMALE RJ21 CONNECTOR
PORT 1	P1+	5	WHT/BLU	26
	P1-	4	BLU/WHT	1
	P2+	3	WHT/ORG	27
	P2-	6	ORG/WHT	2
	P3+	2	WHT/GRN	28
	P3-	7	GRN/WHT	3
	P4+	8	WHT/BRN	29
	P4-	1	BRN/WHT	4
PORT 2	P1+	5	WHT/BLU	30
	P1-	4	BLU/WHT	5
	P2+	3	WHT/ORG	31
	P2-	6	ORG/WHT	6
	P3+	2	WHT/GRN	32
	P3-	7	GRN/WHT	7
	P4+	8	WHT/BRN	33
	P4-	1	BRN/WHT	8
PORT 3	P1+	5	WHT/BLU	34
	P1-	4	BLU/WHT	9
	P2+	3	WHT/ORG	35
	P2-	6	ORG/WHT	10
	P3+	2	WHT/GRN	36
	P3-	7	GRN/WHT	11
	P4+	8	WHT/BRN	37
	P4-	1	BRN/WHT	12
PORT 4	P1+	5	WHT/BLU	38
	P1-	4	BLU/WHT	13
	P2+	3	WHT/ORG	39
	P2-	6	ORG/WHT	14
	P3+	2	WHT/GRN	40
	P3-	7	GRN/WHT	15
	P4+	8	WHT/BRN	41
	P4-	1	BRN/WHT	16

	AM-90 BOARD PAIR #	RJ-45		FEMALE RJ21 CONNECTOR	FEMALE DB9
PORT 5	P1+	5	WHT/BLU	42	
	P1-	4	BLU/WHT	17	
	P2+	3	WHT/ORG	43	
	P2-	6	ORG/WHT	18	
	P3+	2	WHT/GRN	44	
	P3-	7	GRN/WHT	19	
	P4+	8	WHT/BRN	45	
	P4-	1	BRN/WHT	20	
PORT 6	P1+	5	WHT/BLU	46	
	P1-	4	BLU/WHT	21	
	P2+	3	WHT/ORG	47	
	P2-	6	ORG/WHT	22	
	P3+	2	WHT/GRN	48	
	P3-	7	GRN/WHT	23	
	P4+	8	WHT/BRN	49	
	P4-	1	BRN/WHT	24	
PORT 7	P1+	5	WHT/BLU		2 GRN
	P1-	4	BLU/WHT		5 RED
	P2+	3	WHT/ORG		3 BLK
	P2-	6	ORG/WHT		9 YEL
	P3+	2	WHT/GRN		4 ORG
	P3-	7	GRN/WHT		7 BRN
	P4+	8	WHT/BRN		8 SLT
	P4-	1	BRN/WHT		1 BLU
PORT 8	P1+	5	WHT/BLU		2 GRN
	P1-	4	BLU/WHT		5 RED
	P2+	3	WHT/ORG		3 BLK
	P2-	6	ORG/WHT		9 YEL
	P3+	2	WHT/GRN		4 ORG
	P3-	7	GRN/WHT		7 BRN
	P4+	8	WHT/BRN		8 SLT
	P4-	1	BRN/WHT		1 BLU

Figure 3-4: PDB-00359-52 External Cabling Kit

PDB-00359-53

Use this kit to connect a modem to the Eagle 450 with full handshaking support.

AM-90 BOARD			PARALLEL PATCH CORD	MALE DB25 ADAPTER = MODEM			
SIGNAL	PAIR #	RJ-45		RJ-45		DB25 PINOUTS	SIGNAL
RXD	P1+	5	WHT/BLU	5	GRN	3	TXD
RTS	P1-	4	BLU/WHT	4	RED	4	CTS
TXD	P2+	3	WHT/ORG	3	BLK	2	RXD
DTR	P2-	6	ORG/WHT	6	YEL	20	DTR
CTS	P3+	2	WHT/GRN	2	ORG	5	RTS
GND	P3-	7	GRN/WHT	7	BRN	7	GND
DCD	P4+	8	WHT/BRN	8	SLT/WHT	8	DCD
SHLD	P4-	1	BRN/WHT	1	BLU	1	SHLD

Figure 3-5: PDB-00359-53 External Cabling Kit

PDB-00359-54

Use this kit to connect all terminals and printers to the Eagle 450 AM-90 ports.

AM-90 BOARD			PARALLEL PATCH CORD	MALE DB25 ADAPTER = CRTs/PRINTERS			
SIGNAL	PAIR #	RJ-45		RJ-45		DB25 PINOUTS	SIGNAL
RXD	P1+	5	WHT/BLU	5	GRN	2	TXD
RTS	P1-	4	BLU/WHT	4	RED	5	CTS
TXD	P2+	3	WHT/ORG	3	BLK	3	RXD
DTR	P2-	6	ORG/WHT	6	YEL	8	DCD
CTS	P3+	2	WHT/GRN	2	ORG	20	DTR *
GND	P3-	7	GRN/WHT	7	BRN	7	GND
DCD	P4+	8	WHT/BRN	8	SLT/WHT	4	RTS
SHLD	P4-	1	BRN/WHT	1	BLU	1	SHLD

* CRT or printer busy

Figure 3-6: PDB-00359-54 External Cabling Kit

Both of the above cable kits require a shielded parallel patch cord with RJ-45 connectors at each end, like the 10 foot PRA-00189-10 cables supplied in the PDB-00359-56 cabling kit.

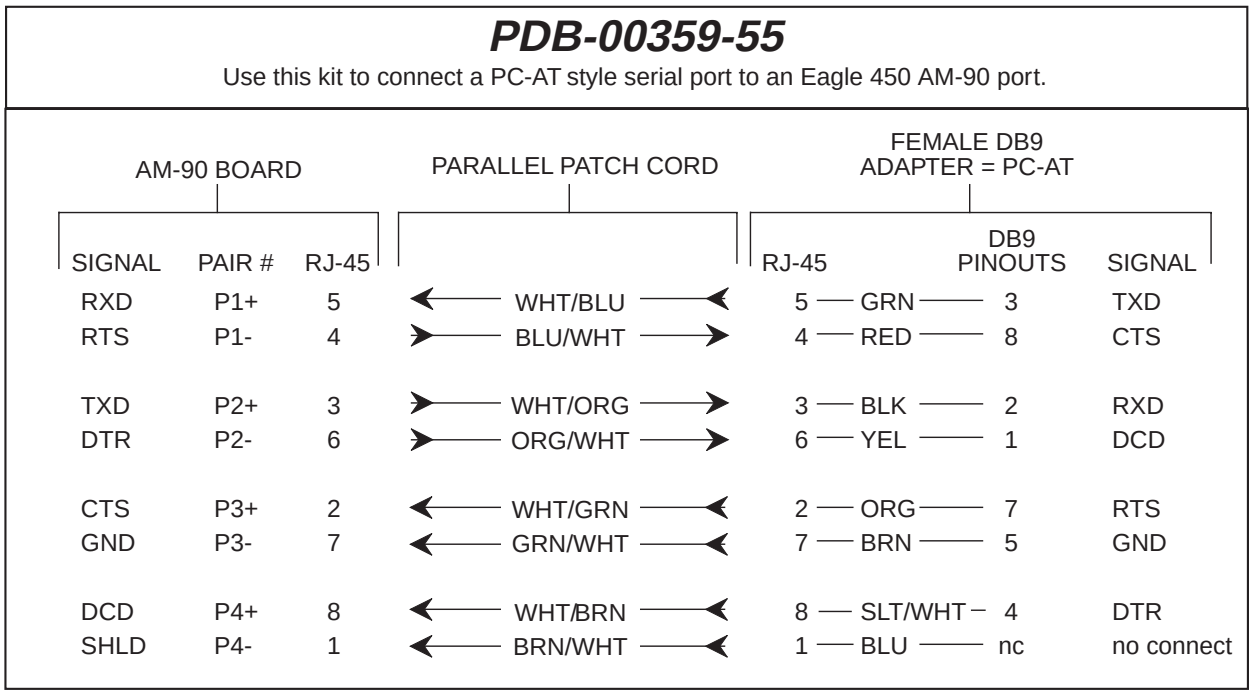

Figure 3-7: PDB-00359-55 External Cabling Kit

The above cable kit requires a shielded parallel patch cord with RJ-45 connectors at each end, like the 10 foot PRA-00189-10 cables supplied in the PDB-00359-56 cabling kit.

PARALLEL I/O SUPPORT

The Eagle 250 includes one parallel port, with a 25-pin connector on the back panel. This port supports the industry-standard Centronics interface. The signal pinouts for the parallel port are as follows:

Pin #	Signal Name	Pin #	Signal Name
1	Data strobe	14	Auto line-feed (not used)
2	Data 1	15	Error
3	Data 2	16	Printer reset
4	Data 3	17	Select in
5	Data 4	18	Ground
6	Data 5	19	Ground
7	Data 6	20	Ground
8	Data 7	21	Ground
9	Data 8	22	Ground
10	Acknowledge	23	Ground
11	Busy	24	Ground
12	Paper error	25	Ground
13	Select		

For best reliability and printer performance, your parallel printer cable should be no longer than six feet.

NETWORK CONFIGURATION

The Eagle 250 includes as standard an RJ-45 10BaseT Ethernet connector attached to the on-board Ethernet controller. You can use this connector to attach the Eagle 250 to your network, then use AlphaTCP for industry-standard TCP/IP networking on your LAN or the Internet.

10BaseT Cabling

10BaseT connections use 100 ohm unshielded twisted pairs, with at least two pairs per cable (one set of pairs for transmitting data and another for receiving). 10BaseT cables terminate in eight pin RJ-45 connectors with the following pin assignments:

RJ-45 Pin Number	Signal Name
1	Transmit Data +
2	Transmit Data -
3	Receive Data +
6	Receive Data -

The 10BaseT specification allows a maximum distance of 100m (approx. 300 feet) between the computer and hub.

EAGLE 250 UPS STATUS PORT

The UPS status port on the back panel of the Eagle 250 is a male DB-9 connector. The UPS system also has a male DB-9 connector for its switch contact port. To connect the UPS to the computer you need a cable with two female DB-9 connectors. The following table shows the pinout connections required to make this cable. You need an 8-wire cable, and both grounds are necessary!

Signal Name	DB-9 Pin # (CPU End)	DB-9 Pin # UPS End
FAULT	3	1
GND	2	2
GND	5	5
ON BYPASS	6	6
LOW BATTERY	7	7
INVERTER ON	8	8
AC PWR FAIL	9	9

Chapter 4 - Installation

This chapter describes what you have to do to get the Eagle 250 up and running. It covers:

- Site requirements and preparation
- Unpacking and preparing the computer
- Initial setup and testing
- Setting CMOS options
- Modifying the initialization file
- Adding user names

SITE REQUIREMENTS

Like any other computer, the Eagle 250 requires an appropriate site. When choosing where to place the computer, consider these conditions:

- Size: Make sure the computer and any other equipment, such as external peripherals, terminals, or printers, will fit with enough room for cables, ventilation, workspace, etc. There must be at least six inches of space behind the main cabinet for ventilation, and cable connections could need even more.
- Power: The computer must have enough reliable, properly regulated electrical power. It may be a good idea to test the line voltage. Many power companies will install test equipment to determine if you need additional line regulation. You can also test line voltages using a high-speed line transient recorder. If, over several days, the line voltage varies more than 10 percent from the rated line voltage, you may need a power conditioner and a new, dedicated AC power circuit.

  To avoid electrical interference, sources of electrical noise such as air conditioners, copiers, or cleaning equipment should not be connected to the same power circuit as the computer.

  There must be enough outlets for the computer and any peripherals. These outlets should have a common grounding point restricted to only those connections coming from the computer installation. If you must use extension cords, be sure the cords are rated for the full amount of current the computer or peripheral requires. For the main computer, the cord rating should be at least 15 amps.

  Power requirements are listed in the Specification chapter, on page 2-1.
- Static: Static electricity can cause the computer to reboot unexpectedly, and can even damage or destroy internal components. Ideally, the computer should not be in a carpeted area. If it must be, use anti-static spray as necessary or install an air conditioner which controls humidity to reduce static electricity.
- Temperature and humidity: The computer is fairly tolerant, but extreme conditions can keep it from working properly. Temperature and humidity limits are given in the Specifications chapter, on page 2-1.

- Cabling: The RS-232 cables to any serial terminals and printers should generally not be longer than 50 feet, and, to avoid interference, should not be too near telephone cables or high-voltage lines. They also shouldn't be in elevator shafts or cross walkways.

  Parallel printer cables normally should be six feet or shorter. External SCSI cables also have length restrictions, as discussed in Chapter 3. If you have printers or external SCSI devices, make sure the places for them are close enough to the main cabinet.

- Cleanliness: Try to locate the computer out of harm's way, where no one is likely to spill food or drink into it, or use its top as a spare table. Dust and dirt can clog the fans and cause heat problems, so keep the computer in a clean, low-traffic area if possible.

UNPACKING AND PREPARING THE EAGLE 250

Unpack the computer and ***save all packing material and cartons***. When re-shipping or otherwise transporting your computer, you must use the original packaging to ensure safe shipment.

Make sure you have all the equipment you ordered: computer, terminals, cables, and so on. If anything is missing, or if it appears any equipment was damaged during shipment, please contact Alpha Micro Sales Order Administration.

Before turning on the computer for the first time, we suggest you open the chassis and make sure all boards are properly seated and cables are securely attached. While the computer was tested at the factory, rough handling during shipment can vibrate connectors loose and keep the system from working.

Checking the AC Voltage Setting

Before plugging in the computer for the first time, check the voltage switch on the back panel. Make sure it is set correctly to 115 or 230 Volts. Turning on the computer with the voltage switch on the wrong setting can seriously damage the computer.

The voltage switch is next to the power cord plug, as shown on the picture of the Eagle 250 back panel in Chapter 3.

Recording the SSD Identification Number

The SSD for the AM-138 board in the Eagle 250 is written on the system ID label on the back of the computer. Please enter the SSD on the Contact Information Sheet in the *Eagle 250 Owner's Manual* so the customer has it easily available if needed.

INITIAL SETUP AND TESTING

Before connecting the entire installation, it's a good idea to test the main computer by itself and with only the operator terminal attached. So, follow this procedure:

1. Again, check the voltage switch on the back panel to make sure it's set correctly.
2. Attach the operator terminal to serial port 0 on the back panel. See the back panel picture in Chapter 3 for the port location. By default, port 0 is set to an Alpha Micro terminal at 19.2K baud.
3. Plug the computer in.
4. To start the self test, hold down the Reset button on the front panel while you press the Power button to turn the computer on. If you aren't familiar with the front panel of the Eagle 250, see the *Eagle 250 Computer Owner's Manual*.

   See the *System Self Test User's Guide* for information on running the self test and interpreting its results. If the computer doesn't come on, the self test doesn't start, or the self test finds a problem, see Chapter 5, "Troubleshooting."
5. After the self test has completed a full cycle of test, press Reset to end the test and boot the computer.

Do not press Reset to end self test during the disk test portion. Doing so could damage the data on the disk!

The computer should now boot using the default initialization file. Assuming it does so successfully, you can continue by modifying the CMOS settings and/or the system initialization file.

MODIFYING THE CMOS SETTINGS

When booting, the Eagle 250 uses data stored in its CMOS parameters to find the primary and alternate boot devices, the system monitor and initialization files to use, and other system options. The CMOS RAM is on the AM-138 board; it is battery backed up and write-protected for data integrity.

You can change the CMOS configuration even if you can't boot the computer under AMOS. This can be very useful in case of certain system problems. To change the CMOS configuration, you must have a terminal attached to port 0 on the AM-138 system board. The first time you access CMOS, this terminal must be set to 19.2K baud; you can then change the terminal speed setting if you want.

It's a good idea to write down the CMOS settings. If you need to replace the batteries, as described in the *Eagle 250 Service Manual*, the settings are lost and must be re-entered.

Eagle 250 Initialization Routine

Before relying on the CMOS parameters, the AM-138 boot code checks the validity of the CMOS contents by verifying the parameter checksum. If the checksum verifies, the system front panel displays "C5" while it verifies CMOS, and boots using the current CMOS parameters. If you want to change the CMOS parameters, press ESC when the "C5" displays (you have approximately three seconds). This displays the CMOS configuration menu, as described below.

If the CMOS checksum test fails, the front panel will blink "CE" for several seconds, then switch to "CC" while the boot code tests CMOS RAM to ensure that it is working. If the RAM test fails, the boot code displays "CF" on the front panel and the system tries to boot using the default settings shown below. If this is the case, contact Alpha Micro's Technical Assistance Center.

CMOS Menu Options

To enter CMOS Setup, first make sure no one else is using the computer. Then, press the Reset button. When "C5" appears on the front display panel, press ESC on the boot terminal to interrupt the boot process and access the CMOS Configuration menu. You have about three seconds to press ESC.

The port 0 terminal’s baud rate ***must*** match the current CMOS setting (default is 19.2K), or CMOS will not recognize the ESC character.

The CMOS menu looks like this:

```
                ALPHA MICROSYSTEMS
          AM-138 CMOS Configuration Menu

   Primary boot device type . . . . . .  SCSI Disk
   Primary boot device unit # . . . . .  0

   Alternate boot device type . . . . .  Streamer
   Alternate boot device unit # . . . .  3

   Boot monitor file name . . . . . . .  AMOS32.MON
   Boot initialization file name  . . .  AMOS32.INI

   Network interface type . . . . . . .  TPI
   Serial port 0 speed  . . . . . . . . .19,200
   Display console boot messages  . . . .Yes

   Use [UP] and [DOWN] keys to select an item.
   Use [LEFT] and [RIGHT] keys to change item.
            Press [ESCAPE] when done.
```

As indicated, you use the ↑ and ↓ keys to select a parameter. To change a parameter, use the ← and → keys to cycle through its possible settings. The only exceptions are file names, which you type.

The CMOS menu fields are:

Primary Boot Device Type

This is the type of device to boot from if the attempt to boot from the alternate device fails or no alternate device is selected. For the Eagle 250, this should always be set to SCSI Disk; do not select Flash.

Primary Boot Device Unit

This sets which primary drive number to boot from. Valid unit numbers are 0-6 and 8-15 (only 0-6 when using the narrow SCSI bus).

Alternate Boot Device Type

This is the type of device to attempt to boot from first, before using the primary boot device. Currently supported alternate boot devices are Streamer, SCSI Disk (which includes the SCSI floppy disk), or None (boot from primary device only). You should not choose Flash or Xmodem on the Eagle 250.

Alternate Boot Device Unit

This selects which alternate device number to boot from. Valid unit numbers are 0-6 and 8-15 (only 0-6 when using the narrow SCSI bus). You should know how many devices are attached to the system before you enter this number. If you select a number and the device is not found, the system will boot from the primary boot selection.

Boot Monitor File Name

Type the name of the monitor file to be loaded during boot. Any valid file name, with an .MON extension, can be used. The monitor file must exist in account [1,4] of the first logical disk of the selected boot device. This parameter is not used when booting from a tape device. You can use the backspace and the arrow keys to edit this field.

Boot Initialization File Name

Type the name of the system initialization (INI) file to be used during boot. Any valid file name, with an .INI extension, can be used. The INI file must exist in account [1,4] of the first logical disk of the selected boot device. You can use the backspace and the arrow keys to edit this field.

Network Interface Type

Always select TPI for the Eagle 250.

Serial Port 0 Speed

By default, CMOS requires a port 0 terminal set to 19.2K baud. You can change this to 1200, 9600, or 38.4K. The speed you set here should match the port 0 definition in the system initialization file. If the terminal's baud rate does not match this setting, you won't be able to re-enter the CMOS menu, since CMOS will not recognize it when you press ESC.

Display Console Boot Messages

When set to Yes, this option displays status messages on the operator terminal during booting. These messages are equivalent to each of the front panel status codes normally displayed during booting, and are normally only needed if you cannot see the status display.

Saving the CMOS Settings

When you are finished making changes press [ESC]. A message will appear at the bottom of the screen asking if you wish to save any changes made. Type **Y** to save the changes in the CMOS RAM, or **N** to abandon any changes made. After you enter your response, the system will boot using the new parameters, if you saved them.

Important Note

If you want to boot from a physical disk device other than device ID 0, you should create a disk driver for the selected drive ID and MONGEN it into the monitor. ***You must do this if you want to be able to MONTST using your boot monitor***. While a hardware reset will work if the monitor contains the generic SCZ138.DVR, because it reads the drive ID from CMOS, MONTST does not look at the CMOS settings, and so will not know which drive to boot from unless the drive ID is embedded in the driver.

MODIFYING THE INITIALIZATION FILE

The modifications you'll need to make to the default system initialization command file for the Eagle 250 are similar to the modifications for other AMOS computers. You'll probably want to set the number of jobs, add programs to system memory, set up AlphaTCP networking, define terminals and printers, and so on. You may also want to change SCSI dispatcher options, set read-ahead, or disable Super I/O for some or all serial ports.

The name of the system initialization file for the Eagle 250 is usually AMOS32.INI (though, because of the CMOS feature, it can have any name you want).

NEVER change the system initialization file directly! Always make a copy of it and modify the copy. While the ability to change the initialization file name through CMOS can make it possible to boot even if you invalidate the normal initialization file, it is still safer to work in a copy of the file.

To make a copy of the system initialization file to modify and test:

1. Log into DSK0:[1,4] and copy the file:

```
LOG DSK0:[1,4] [ENTER]
COPY TEST.INI=AMOS32.INI [ENTER]
```

2. Use AlphaVUE or another text editor to edit the contents of the test file:

```
VUE TEST.INI [ENTER]
```

3. Make all the necessary changes to TEST.INI. Save the file when exiting AlphaVUE by pressing the [ESC] key and typing **F**. Some of the common changes you'll make are discussed below.

4. Make sure no one else is using the computer and use MONTST to insure TEST.INI works as expected:

   LOG OPR: ENTER
   MONTST TEST.INI ENTER

5. After you have successfully tested TEST.INI and you are satisfied with the results, copy it back to AMOS32.INI.

Be careful not to rename the TEST.INI file too soon. You might want to let the computer run awhile to test out the new configuration before you rename it. If you decide you don't want to keep the new configuration, you can always press the Reset button to reboot with your former system initialization file.

Follow this procedure any time you modify the system initialization file.

Do not change any lines in the system initialization file unless you're familiar with them and understand their ramifications. For more information on the initialization file, refer to the *System Operator's Guide to the System Initialization Command File, DSO-00002-00.*

```
:T
;
JOBS 5
;
JOBALC JOB1
;
QUEUE 2000
;
TRMDEF TERM1,A31810=0:19200,ALPHA,200,200,200,EDITOR=15
;
PARITY                          ; Clear memory
VER                             ; Unlock keyboard
;
SCZDSP SCZ138.SYS/ET            ; SCSI dispatcher
;
DEVTBL DSK
DEVTBL TRM,RES,MEM
DEVTBL /STR0                    ; Streaming tape device
;
BITMAP DSK                      ; Paged bitmaps for AMOS 2.X
;
ERSATZ ERSATZ.NEW
MSGINI 20K
;
SYSTEM SYSMSG.USA
SYSTEM DCACHE.SYS/N/M 1M        ; Enable disk read-caching
SYSTEM DVR:DSK/N 100K 60        ; Enable disk write-caching
SYSTEM CMDLIN.SYS
SYSTEM SCNWLD.SYS
SYSTEM QFLOCK.SYS
SYSTEM TRM.DVR[1,6]
SYSTEM STR.DVR[1,6]
SYSTEM EGP.DVR[1,6]
SYSTEM
;
SET DSKERR
;
MOUNT DSK:
;
MEMORY 0
```

Simple Initialization File

Defining the Parallel Port

The Eagle 250 contains one parallel port, which is not automatically defined for you. To enable the parallel port:

1. Type the following statement following the last DEVTBL statement in your TEST.INI file:

   ```
   DEVTBL /EGP0
   ```

2. Load the parallel printer driver into system memory using a SYSTEM statement:

   ```
   SYSTEM EGP.DVR[1,6]
   ```

3. To connect a printer to the parallel port, change the DEVICE statement in the printer initialization file to reference parallel port 0 (EGP0:). For example:

```
DEVICE=EGP0:
```

For information on printer initialization files and printer spoolers, see the *System Operator's Guide*. For cable pinouts for a parallel printer, see Chapter 3.

Defining Ports and Jobs

When the Eagle 250 leaves Alpha Micro, only the operator terminal is defined in the initialization file. After installation, you'll want to define any other terminals and serial printers. To do this, you need to:

- Define additional jobs
- Define the terminal or printer connected to each serial port
- Attach the appropriate job to each defined terminal or printer.

Before adding jobs to the system, make sure its AMOS license allows the number of jobs you're defining.

The steps below outline the process of defining the terminals and printers:

1. Make a copy of the initialization file, as discussed previously.
2. At the beginning of the TEST.INI file is a JOBS statement. The number following the JOBS statement represents the total number of jobs on the computer. Change the number in the JOBS statement to the number of jobs needed. For example:

```
JOBS 17
```

When calculating the number of jobs, be sure to include all the jobs the computer will use: background jobs, printer jobs, network connections, and so on. It's often a good idea to add a few extra when possible.

3. On the lines following the JOBS statement, there are one or more JOBALC statements. You can define each job in a separate JOBALC statement, or you can define several jobs in the same JOBALC statement by separating the jobnames with commas. For example:

```
JOBALC JOB4,JOB5
```

Jobnames can have up to six characters (A - Z, 0 - 9, $). Each jobname defines a job on your computer. The total number of jobs defined in JOBALC statements must be equal to or less than the number in the JOBS statement.

4. Next, define the devices attached to the serial ports using TRMDEF statements. Here is a sample TRMDEF statement:

```
TRMDEF TERM2,A31810=1:19200,ALPHA,200,200,200,EDITOR=10
```

Using the sample, here's the information you must tell AMOS about a terminal:

- TERM2 is the terminal name. You may use any name containing six or fewer letters and/or numbers. Each terminal must have a different name.

- A31810 is the name of the interface driver (without the .IDV extension) for the circuit board the terminal is connected to. The ports on an Eagle 250 use these interface drivers:
  - A31810: The four or eight serial ports on the AM-138 system board
  - A31810: Ports on an AM-318-10 board
  - AM318: Ports on an AM-318-00 or -02 board
  - AM314: Ports on an AM-314 board
- 1 is the number of the port the terminal is attached to. The port number is octal, and depends on the port the terminal is attached to on the I/O board, and the I/O expansion slot the board is in. This table gives the port numbers for each board type/slot combination:

Expansion Slot	J7	J8	J9
Any AM-318 (-00, -02, or -10)	10-17	20-27	30-37
AM-314	10-13	20-23	30-33

  Use these numbers even if it means you skip some numbers. For example, if you have an AM-314 board in J7 and an AM-318-10 in J8, the AM-318-10 ports are numbered from 20-27, and numbers 14-17 are not used.

  The eight ports on the AM-138 board are numbered from 0-7.
- 19200 is the baud rate of the terminal.
- ALPHA is the name of the terminal driver. ALPHA can be used for most Alpha Micro terminals; you can change this to a driver specifically for the terminal if one exists.
- 200,200,200 are buffer sizes, expressed in number of characters. Different situations may require other buffer sizes, but this is typical.
- EDITOR=15 enables the line editor for this terminal. See the *AMOS User's Guide* for more information on the line editor.

5. Use a SETJOB statement for each terminal defined in a TRMDEF statement to link it to a job defined in a JOBALC statement. You can include parameters in the SETJOB statement to:
   - Define how much memory to allocate to a specific job.
   - Define the specific disk and account you want the specific job to log into each time the computer boots.
   - Unlock the terminal’s keyboard (with the VER command).

   SETJOB statements must be after the last SYSTEM statement in the .INI file. Here is a sample:

   ```
   SETJOB JOB4,TERM2,256K,LOG DEMO,VER
   ```

6. Add a WAIT command to give the computer time to process the commands for the job before it proceeds with the rest of the initialization file.

   ```
   WAIT JOB4
   ```

7. When you have added SETJOB and WAIT statements for each job, perform a test reboot, using the procedure earlier in this chapter.

For more information on defining jobs and terminals, see the *System Operator's Guide to the System Initialization Command File*.

AlphaTCP Setup

If your Eagle 250 is connected to a network, the AlphaTCP networking software allows any network user—and any external user who has access to your network—to use the programs and files on the Eagle 250. To use AlphaTCP, which is included with AMOS on the Eagle 250, you must start various background "server" jobs to handle the network traffic and other tasks, and assign memory for use by people requesting a connection over the network.

The initialization file setup for using the AlphaTCP networking software can be involved, and in addition to changing your initialization file, you need to modify the AlphaTCP configuration files at installation, and possibly when you add new network users.

Configuring your initialization file to make the best use of your resources with AlphaTCP is too lengthy a process to discuss here. Please see your *AlphaTCP Administrator's Guide*, DSO-00187-00, especially chapters 3 and 4, for detailed information and instructions.

When defining the network driver for the Eagle 250, in most situations you should use AM138.LDV for best performance. In certain circumstances—for example, if you need to perform wildcarded AMOS copy commands across the network—use AM138.NDV instead.

Changing User Memory

In addition to the operating system itself, the memory of the Eagle 250 is used by:

- The serial ports on the AM-138 CPU board and any added I/O boards
- Background processes, such as the AlphaTCP server programs
- People connecting to the computer over the network

Whenever you change any memory allocation, you need to balance all three types of requirements.

You can display the current memory allocations on your computer using the STAT or SYSTAT command. Neither STAT nor SYSTAT show how much memory is left in the SMEM memory pool.

The amount of memory for each serial port and background process is set in a SETJOB statement in the initialization file (except for serial port 0, which usually receives all of the memory not used for any other purpose). Each network user, however, may be able to request a specific amount of memory when he or she connects to the computer (depending on the AlphaTCP configuration, network users may be assigned a memory size automatically). So, rather than assigning memory individually to each job, the initialization file merely allocates a "pool" of memory for network jobs. When someone connects to the Eagle 250, memory is taken from this pool and assigned to that network user; when the user disconnects, the memory is returned to the pool.

The size of this memory pool is set by an SMEM (shared memory) statement in the initialization file. The memory pool allocated by SMEM is also used for some other purposes, such as spawned MULTI jobs, but for this discussion we'll consider that it's all available for network users.

If you decide to change the amount of memory allocated to a job, or the memory pool, you can modify the system initialization file following the instructions below.

Be very careful when changing memory allocations. If your Eagle 250 uses AlphaTCP networking, most of the memory could be used for AlphaTCP processes and users connected via the network. Allocating memory incorrectly could make it impossible for some or all network users to access the computer. Also, changing the memory for AlphaTCP processes or available for network users may require changes to one or more of the AlphaTCP configuration files in addition to the initialization file. Be sure to refer to the *AlphaTCP Administrator's Guide*, DSO-00187-00, before changing any aspect of AlphaTCP's memory use.

Also, please refer to the *System Operator's Guide to the System Initialization Command File* for more information on SETJOB and SMEM.

1. Make a copy of the initialization file and edit it, as discussed previously.
2. To change the memory for a serial port or a background process, such as the AlphaTCP TCPEMU server, scan through the file until you come to the SETJOB statement for that job. The amount of memory for the job is immediately after the terminal name in the SETJOB statement. You can increase or decrease this amount, as long as you keep in mind how much total memory your computer has, and how much memory the other jobs, and the shared memory pool, need.
3. To change the amount of memory in the shared memory pool, find the SMEM statement; it should be just after the last SYSTEM statement. Again, you can increase or decrease this amount, but be sure to take all of your users' memory needs into account.

   Remember, the operator terminal—the one the computer boots on, connected to port #0—usually gets all the memory not otherwise used. To increase the memory for this job, you need to decrease memory use somewhere else; to decrease it, you can add memory to another job or the memory pool.

   You can assign a particular amount of memory to port #0 by using a FORCE command. If you do this, any memory not assigned to any job or process is wasted.

4. When you've made your changes, save the TEST.INI file and exit AlphaVUE. Then, perform a test reboot by running MONTST as described earlier in this chapter. You can then use the STAT or SYSTAT command to see how much memory is assigned to each job.

SCSI Dispatcher Options

Like other computers in the Eagle family, the Eagle 250 uses a program called the SCSI dispatcher to handle SCSI bus communication. The dispatcher for the Eagle 250 is called SCZ138.SYS. The provided initialization file includes the command SCZDSP SCZ138.SYS to start the dispatcher. You may want to add one or more of these options to that line:

Switch	Description
/ET	Enable tolerant active negation
/EW /EW:{id#}	Enable Wide SCSI negotiation for all devices Enable Wide SCSI negotiation for SCSI device ***ID#***
/NQ /NQ:{id#}	Disable Command Queuing for all devices Disable Command Queuing for SCSI device ***ID#***
/NS /NS:{id#}	Disable Synchronous Negotiation for all devices Disable Synchronous Negotiation for SCSI device ***ID#***
/NP	Disable Parity Checking (Parity still generated) on all devices

We recommend that all Eagle 250s use the /ET switch to enable tolerant active negation. Using active negation improves bus integrity when using synchronous data transfers.

Never attach a device to or remove a device from the SCSI bus while system power is on. This is never a good idea, but active negation makes it even more likely that doing so will damage the device, the SCSI controller, or both.

The most common of the other switches is /EW, to enable Wide SCSI operation when using the optional Wide SCSI bus. For example, if you are using the Wide SCSI bus and have a Wide SCSI disk drive at ID 0, with narrow devices at other IDs, use this statement to enable Wide SCSI operation for just that drive:

```
SCZDSP SCZ138/ET/EW:0
```

If you have both Wide and narrow SCSI devices attached to the Wide bus, enable Wide SCSI operation only for the Wide devices. Use /EW without a device ID (to enable Wide operation for the entire bus) if all devices on the bus, both disk and tape drives, are Wide SCSI devices.

Do not use the /EW switch with the narrow SCSI bus, even if you have Wide SCSI devices attached to the bus using the appropriate adapters.

Setting Write Buffering

To optimize SCSI bus performance, you may want to enable write buffering, in which blocks to be written to the disk are held (buffered) in memory and written in groups, more efficiently than they could be individually. For information on write buffering, including the modifications to make to your initialization file to enable it, please see Appendix A.

Disabling Super I/O

If you want to disable Super I/O on some or all ports, see Appendix B for instructions. Disabling Super I/O will decrease I/O performance; you should generally do this only if an application program will not run on a Super I/O enabled port.

Multiple Initialization Files

The Eagle 250's CMOS menu feature lets you easily change the initialization file the computer boots from. Because of this, we suggest you keep more than one initialization file ready for use. That way, if you run into a problem with one file, you can change the CMOS setting and reboot using a different file that you know is good. You can even have different files for different configurations: your usual file with the network enabled and all ports defined, a simple single-user file, and possibly others.

ADDING USER NAMES

All users, whether on a serial terminal or a networked workstation, need to enter a user name (and possibly a password) to log on to the Eagle 250. As the computer is shipped from Alpha Micro, two user names are defined, `Demo` and `System Service`. You will no doubt want to add more names for your users. You can do so with the MUSER command.

You can use MUSER only from OPR:

```
LOG OPR: [ENTER]
MUSER [ENTER]
```

At the MUSER menu, type **L** [ENTER] to list the currently defined users, or **A** [ENTER] to add new users. Besides name and password, you can define many characteristics and limits for each user. For details, see the MUSER sheet in the *AMOS System Commands Reference Manual*.

Chapter 5 - Troubleshooting

As with previous Alpha Micro models, we expect the great majority of Eagle 250 installations to go smoothly, and the computers to work properly and reliably. However, there will be instances where something doesn't work correctly after installation, or problems arise sometime afterward. When this happens, what do you do?

The number of variables involved in isolating and correcting a problem with a computer installation makes it almost impossible to give specific instructions which will apply to even a majority of situations. So, instead, this chapter discusses some general troubleshooting procedures and strategies which could help you diagnose and fix a wide range of possible problems.

The *Eagle 250 Owner's Manual* contains instructions for the end user on what he or she should do in the event of a problem. This includes giving you, the dealer, detailed information about who was affected, error messages and other symptoms, and what each user was doing when the problem occurred. Getting complete and accurate information from the problem site is the first step toward fixing the problem. This discussion assumes you have worked with the end user to get this information

Once you have information about the error or other system problem, here are some possible ways to find and correct it. Not all of these strategies will apply to any specific problem, and you won't necessarily perform them in the order they are given here. Your experience with Alpha Micro computers and your application software can probably get you started in the right direction.

Some of these procedures require you to open the computer chassis and check or change components. ***This should be done only by qualified service personnel;*** the high voltage present creates a risk of electric shock.

- If a status code (anything other than 0) appears on the front panel, check the list later in this chapter to see what it indicates.
- If the problem occurs more than once, look for patterns: what was happening when the problem occurred that's different than at other times. This could be almost anything: a particular program or combination of programs being used, particularly heavy use of the computer or the network, even a cleaning crew vacuuming near the computer, causing static electricity.
- Especially if the computer won't boot, run the system self test to check the operation of the hardware involved in booting. See the *System Self Test User's Guide*, DSO-00156-00.
- Make sure all software and hardware in the computer is up to the latest revision. Check the patch area of TABBs (Alpha Micro's Technical Assistance Bulletin Board) to see if there are any patches outstanding for any software that may be involved, especially any that describe a problem similar to the one you're trying to solve.
- If any hardware or software was recently added to the system, it's an obvious suspect. Make sure all programs and hardware components are mutually compatible. Try removing the recent addition (or returning to the previous version) and see if the problem goes away.
- If the configuration was recently changed (for example, you changed the files placed in system memory during booting), return the system to its previous configuration.

- If you can, try to duplicate the problem conditions on another computer in your office. If you can create the same failure, you can then attempt to find and fix it without disrupting your customer's daily routine or reconfiguring their computer.
- If it looks like the problem is related to a specific piece of hardware, either remove that component from the computer (if it's optional), or replace it with a known good component.
- Try turning off or changing software and/or hardware options that may be contributing to the problem. For example, you could change the number of read-ahead blocks in a disk driver, turn off (or on) write caching, etc.

Finally, remember that we're always here to help. Just call the Alpha Micro Technical Assistance Center at 800/487-7787.

FRONT PANEL STATUS DISPLAY CODES

During normal operation, the two-digit display on the front panel displays a zero. When the computer boots, a series of codes appears in rapid succession on the display as the AMOS operating system gets itself up and running. If an error occurs during booting, one of these codes may remain on the status display. This can tell you what was happening when the error occurred.

Some of these codes may also occur any time the computer is running. If you have a problem with the computer, check the front panel; if anything other than 0 is displayed, look it up in this table to help diagnose the problem.

Front Panel Status Code	Description
0	Computer is functioning normally.
4	System has gone under 100 queue blocks remaining. This code displays until replaced by another, even if the system now has over 100 queue blocks. At a convenient time, allocate more system queue blocks in the system initialization file.
7	LEVEL7 debugger is active. Will occur only when the LEVEL7 software is loaded in system memory. Refer to the LEVEL7 documentation for details.
Steady 8	The computer's internal DC power supply has detected an internal power failure.
Flashing 8	UPS low battery condition has existed for more than two minutes. The system has executed a shutdown procedure. Write-caching is deactivated on any disk drives which had it enabled. The system is now in a HALT condition until the batteries in the UPS are completely dead. Turn off system power! To reset this condition, you must get the UPS primary power back on line. Check the UPS input power breaker and be sure it's ON! The system must be rebooted with a hardware reset.
9	Memory parity error was encountered.
F	During booting, memory is being cleared and sized.

Front Panel Status Code	Description
10	An interface driver (.IDV) defined in a TRMDEF statement in the system initialization file was not found in account [1,6] on first logical of the boot device.
11	A terminal driver (.TDV) defined in a TRMDEF statement in the system initialization file was not found in account [1,6] on the boot device.
12	AMOS system initialization (.INI) file not found in account [1,4] on the first logical of the boot device.
20	The computer is beginning to execute the boot PROM. An error at this point indicates your computer has a faulty PROM. Contact your VAR.
21	The computer is transferring the instructions from the PROM into its Random Access Memory (RAM). If an error occurs here, the computer might have a bad PROM or bad memory.
22	The computer is generating a checksum of the instructions in Random Access Memory. If this calculated checksum does not match the checksum coded into the instructions themselves, you see a 2E error code. If an error occurs here, the computer might have a bad PROM or bad memory.
24	Reading the Master File Directory (MFD) from disk. An error at this point indicates disk problems in the alternate boot device.
25	Searching for the User File Directory (UFD) for account [1,2] on the first logical of the alternate boot device.
28	Searching for account [1,4] on the first logical of the alternate boot device. An error at this point may indicate disk problems. Reload the latest version of the system software.
29	If you are ***booting from a tape device***, the computer is searching for a label block on the tape, as the alternate boot device.
2A	Loading the AMOS monitor from the alternate boot device. If an error occurs at this point, reload the latest version of the system software.
2b	Beginning to execute the AMOS monitor program from the alternate boot device. If an error occurs at this point, reload the latest version of the system software.
2d	Bootup from the alternate boot device failed because of a time-out error. This may indicate a faulty memory or an addressing problem.
2E	Bootup from the alternate boot device failed because of a bootstrap loader program checksum error. This may indicate a bad PROM or bad memory.
2F	Bootup from the alternate boot device failed because of an invalid boot device selection. Access the CMOS setup routine by pressing Reset, then pressing the ESC key when the code "C5" appears on the front panel. Verify the device and unit number settings on the CMOS menu.

Front Panel Status Code	Description
33	Initializing the primary boot device. If the boot stops at this point, it may indicate a hardware problem with the primary boot device. When booting from a disk, this code may remain on the display for a short time after you've turned the power on, while the disk drive spins up to operating speed.
34	***While booting:*** reading the Master File Directory (MFD) from the primary boot device. An error at this point indicates disk problems. ***During normal operation:*** unimplemented integer instruction encountered.
35	Searching for the User File Directory (UFD) for account [1,2] on first logical of the primary boot device.
38	Searching for account [1,4] on the first logical of the primary boot device. An error at this point may indicate disk problems. Reload the latest version of the system software.
39	The computer is looking for the system monitor file in [1,4] of the primary boot device. If this file is missing, reload the latest version of the system software.
3A	Loading the AMOS monitor from the primary boot device. If an error occurs at this point, reload the latest version of the system software.
3b	Beginning to execute the AMOS monitor from the primary boot device. If an error occurs at this point, reload the latest version of the system software.
3F	Bootup failed because of an invalid primary boot device selection. Access the CMOS setup routine by pressing Reset , then pressing the ESC key when the code "C5" appears on the front panel. Verify the device and unit number settings on the CMOS menu.
40	UPS inverter on. Normal operation recovering from an AC power failure. Only displays if the UPS status cable is installed on the UPS port.
41	UPS is bypassed. System running without UPS battery protection. Please turn on UPS run switch.
42	UPS inverter on, AC power failure detected, and the system is running off batteries. This status is updated every second that the AC power is off.
43	Not valid, system inoperative.
44	UPS is on, and batteries are low. Batteries are recharging, please check.
45	UPS is bypassed and batteries are low.
46	UPS inverter is running and a low battery condition exists. Only **2 minutes of power remaining!** System shutdown is imminent.
47	Not valid, system dead.
48	An internal fault has been detected in the UPS.

Front Panel Status Code	Description
49	An internal fault has occurred in the UPS and it is in bypass.
4A - 4E	Not valid.
4F	UPS cable has been unplugged from UPS port.
88	Trying to execute an unimplemented SVCA (Monitor) call.
95	Logger CREATE error.
96	Logger CLOSE error.
97	Logger FILOTX error.
98	Logger OPENA error.
99	Logger semaphore not available.
B0	CMOS batteries need to be replaced. Data that was stored in CMOS RAM is not guaranteed.
B1	The battery voltage dropped temporarily (i.e., after replacing the batteries). The initial CMOS parameters are not guaranteed. Access the CMOS setup routine by pressing Reset, then pressing the ESC key when the code "C5" appears on the front panel. Re-enter CMOS settings.
C5	Do you want to enter CMOS setup? You have three seconds to press the ESC key on the boot terminal.
CC	The computer is testing the CMOS RAM for proper operation after detecting a CMOS checksum error.
CE	CMOS parameter checksum error. This code will flash on the front panel for several seconds.
CF	CMOS RAM failure. The system will try to boot using the default CMOS settings shown on page 4-4. Contact the Alpha Micro Technical Assistance Center.
D0	Tried deleting all three valid timer chips and all failed.
EF	System has detected an attempt to write outside of the supervisor stack area.
F	On boot-up, memory is being cleared and sized.
F0	Bus error was encountered.
F1	Address error was encountered.
F2	Illegal instruction was encountered.
F4	CHK instruction.
F5	TRAPV instruction.

Front Panel Status Code	Description
F6	Privilege violation.
F7	TRACE return.
F8	Coprocessor protocol violation.
F9	EM1111.
FA	FPCP branch or set on unordered condition, or FPCP inexact result, or FPCP divide by zero, or FPCP underflow, or FPCP operand error, or FPCP overflow, or FPCP signaling NAN, or FPCP unimplemented data type.

For more information on status display codes, and on how your programs can send a number to the status display, please refer to the *System Operator's Guide*.

Other status codes can appear during Self Test; these codes are discussed in the *System Self Test User's Guide.*

Appendix A - Read-Ahead and Write Buffering

The AM-138 CPU board uses a programmable RISC DMA controller for SCSI bus communications and for data transfer to and from the AM-138's SCSI bus. The Coldfire CPU is only involved with setup before and cleanup after a SCSI command is sent to a device—the rest of the command, including data transfer, is handled by the RISC processor.

Having the RISC processor take care of these details allows both read-ahead and write buffering without the need for a separate controller.

READ-AHEAD

The AM-138's SCSI disk driver, SCZ138.DVR, is able to perform read-ahead directly into the AMOS disk cache (DCACHE). When any program attempts to read a physical block from a disk, the SCZ138 driver will also read up to an additional seven sequential blocks from the disk drive and store these read-ahead blocks in the cache.

This read-ahead scheme works very well when jobs on the system are doing a large number of sequential reads. For example, data base searches and programs like REDALL may execute much faster because the data they require is already in memory and only has to be transferred from the cache into the user partition.

Programs that do significant random disk access (such as RNDRED) tend to slow down with this read-ahead scheme. Most of the slowdown is caused by "thrashing" of the disk cache, where cache entries that will be used again are removed from the cache due to the allocation requirements of the read-ahead blocks (which typically are never used). The actual data transfer overhead is very little, as most SCSI disk drives (especially fast SCSI-2 drives) have a track cache built into the drive allowing both the target and read-ahead blocks to be transferred over the SCSI cable without delay.

Controlling Read-Ahead

For read-ahead to occur on the AM-138 board, both the AMOS disk cache (DCACHE.SYS) and the full SCSI dispatcher (SCZ138.SYS) must be installed. The number of read-ahead blocks to be transferred into cache on every physical disk read is contained in the disk driver. You can set this using the FIXLOG program to generate a disk driver with the appropriate read-ahead blocking factor.

When you use FIXLOG to create a driver for the AM-138 board, you are asked to specify the number of read-ahead blocks. For example, type:

```
FIXLOG ENTER
FIXLOG.LIT Version x.x(xxx)

1. Change the number of logicals.
2. Create a sub-system driver.

Enter choice: 2 ENTER

Enter name of generic driver to be used: SCZ138 ENTER
Enter number of logical units per physical unit: 10 ENTER
Enter SCSI id (0-15): 0 ENTER
Enter number of read-ahead blocks (0-7): 5 ENTER
Enter new driver name: DSK ENTER
New driver is now in memory.
```

To save the driver you have created, type:

```
SAVE DSK.DVR ENTER
```

If you want to disable or change the number of read-ahead blocks, simply use FIXLOG to generate a new disk driver. If the disk driver is for the DSK: device, don't forget to use MONGEN and embed the new driver into the system monitor.

The generic AM-138 SCSI disk driver (SCZ138.DVR) is set up for seven read-ahead blocks.

WRITE BUFFERING

AMOS (and therefore every application written for AMOS) understands only 512-byte disk blocks. Therefore, when a disk write request is made by a program, a single block transfer is made to the disk drive. If the program then writes the next sequential block, the system must wait the latency time of the drive (i.e., the time it takes the drive to complete one revolution) before the next block can be written. Latency, even on fast SCSI-2 drives, is around 7ms.

Write buffering can speed up the write process. When write buffering is enabled, all writes to the SCSI disk are first transferred into a buffer. If the write buffer becomes at least half full, or around three quarters of a second passes with no reads, or if a preset "guaranteed flush" time-out occurs, the SCZ138.DVR disk driver begins scanning through the write buffer, finding blocks that need to be written out to the drive. The algorithm used to flush blocks out to the drive is able to find up to eight consecutive blocks and write them to the disk drive as a single write command, therefore dramatically improving system performance.

Another benefit of write buffering is it tends to eliminate duplicate disk writes, such as bitmap updates during operations such as copying files and tape restores, and prevents head thrashing when reading through random access data files and writing a sequential file out to the disk (as most report generation programs do).

Potential Pitfalls

Obviously, there can be problems with write buffering, especially if the system either crashes or is powered off while writes are pending in the write buffer. If that happens, all pending writes are lost. Though this sounds like a major problem, it can also happen if write buffering is not enabled. However, write buffering increases the number of writes at risk. The primary write buffering risks are an errant software operation or a hardware failure that causes a system crash.

To help reduce the possibility of data loss, certain safeguards have been put in place. Writes are not buffered indefinitely; they are performed whenever the device is not performing reads. Even if the drive is busy with read requests, the buffer is still periodically flushed, based on a user definable "absolute flush time." In addition, if you have a UPS installed and connected to the AM-138 board's UPS status port, and you experience a power failure and the UPS status port senses a low battery condition, AMOS will flush and disable the write buffer in preparation for a system shutdown. Also, the MONTST command automatically flushes the write buffer.

Therefore, you must weigh the potential for data loss (which is always there) versus the dramatic performance increase seen when using write buffering. If you are worried about the reliability of write buffering, it may be worth keeping in mind that the AM-520 disk controller has always used write buffering on a track-by-track basis (however, not quite as efficiently as the AM-138 board’s write buffering scheme). The SMARTDRV program that comes with MS-DOS does write buffering (you may have noticed the "Waiting for system shutdown" message when rebooting a PC with CTRL-ALT-DELETE) and UNIX-based computers have always done it.

Setting Up Write Buffering

To enable write buffering, you must be using the full SCSI dispatcher (SCZ138.SYS). Enable write buffering by adding parameters to the SYSTEM statement used to load the SCSI disk driver into system memory. Append "/N" followed by the buffer size and flush period enables write buffering for that device. The syntax is:

```
SYSTEM DVR:dev/N buffer-size flush-period
```

The ***buffer-size*** is the size of the write buffer (you specify the size in Kilobytes). We advise a buffer size of 100K to 200K.

The ***flush-period*** is the maximum number of seconds data may be left in the write buffer without being written to the disk. For example, if you specify 30, you will know that after 30 seconds any pending writes will be written to the disk. This is true even if the disk is constantly busy servicing reads.

For example:

```
SYSTEM DVR:DSK/N 200K 60
```

One SYSTEM command is required for each different SCSI disk driver present in the system. For example, if you have two 1.2GB SCSI-2 drives named DSK0-36 and DSK37-73 and one 540MB SCSI-2 drive named SUB0-17, you need one SYSTEM command for the DSK device (although it's really two physical drives) and one SYSTEM command for the SUB device.

When specifying write buffering for a device, two files are placed into system memory: .DVR (loaded from disk) and .WRC (directly created in system memory), which are the driver and cache buffer. This is true for all SCSI disk devices except the DSK device. For the DSK device, the file DSK.DVR does not need to be created because it is already loaded into the system monitor. Therefore, for the DSK device, only the file DSK.WRC will be created in system memory.

In the three-drive example mentioned earlier, the added SYSTEM commands would look like this:

```
SYSTEM DVR:DSK/N 100K 60  ;Driver in AMOS will create DSK.WRC
SYSTEM DVR:SUB/N 100K 60  ;Load SUB.DVR and create SUB.WRC
```

This would set up 100K of write buffering for the DSK devices and 100K of write buffering for the SUB device. All three drives would have their write buffers flushed every minute (or sooner if the drives are not busy with read requests).

FINAL NOTES

Both read-ahead and write buffering dramatically improve system performance in our lab tests. Both schemes are fine-tuned for the Coldfire processor and RISC SCSI controller and do not take cycles away from AMOS like other commercially-available disk optimization software.

Although our lab tests attempt to simulate the "real world" of user applications, they probably use the resources of the AM-138 CPU and SCSI subsystem differently than your application does; therefore we highly recommend you experiment with cache and write buffer sizes, read-ahead blocks, and flush periods on an installed system to find the best possible combination for that system.

Appendix B - Super I/O

Super I/O is designed to significantly increase character output for all serial ports using the A31810.IDV or AM318.IDV driver. This includes all AM-318-xx board serial I/O ports, as well as the Eagle 250's eight on-board serial ports. The more terminals you have on your system performing character output, the more you will benefit from Super I/O.

The AM-314 board does not support Super I/O.

Super I/O handles character output in a much more efficient manner than any other previously released AMOS serial port driver, which greatly reduces the load on the CPU and makes more CPU cycles available for other tasks.

In earlier versions of AMOS, Super I/O was a separate, optional feature. In all versions of AMOS supported on the Eagle 250, Super I/O is included in the operating system.

DISABLING SUPER I/O

For most Super I/O-compatible serial ports, you will always want to have Super I/O enabled. However, if you have an application program that does not run properly on a Super I/O-enabled serial port, you can disable Super I/O. You can disable Super I/O on one or more AM-318-xx boards, but you can't disable Super I/O on an individual serial port.

Disabling Super I/O on All I/O Boards

If the application causing problems is one you run infrequently, you can temporarily disable Super I/O on all your AM-318-xx ports by making the following changes to your initialization file:

As with any change to the initialization file, use proper caution—make your changes to a copy of the initialization file, not the file you use to boot the computer.

1. Find the first TRMDEF statement that uses the A31810.IDV interface driver—this is usually port 0. Disable Super I/O by adding /O (that's an "O", not a zero) to the statement. For example, change:

   ```
   TRMDEF TERM1,A31810=0:19200,AM65,100,100,100
   ```

   to

   ```
   TRMDEF TERM1,A31810/O=0:19200,AM65,100,100,100
   ```

2. If any ports use the AM318.IDV interface driver, find the first TRMDEF statement using it. Make the same change to it: add /O after `AM318`.
3. Save the file and MONTST using it to disable Super I/O.

Once you have completed running the program that exhibits problems under Super I/O, you can reboot your computer using the standard system initialization command file, which will reactivate Super I/O.

Disabling Super I/O on Individual Boards

When your computer is processing your system initialization command file, it loads a copy of the A31810.IDV (or AM318.IDV) file when it encounters the first TRMDEF statement using it. This same copy of the .IDV file is used for all subsequent I/O boards. This being the case, if you use the **/O** switch in the first A31810 TRMDEF statement, Super I/O is disabled on all I/O boards using that interface driver. In some cases, you may want to disable Super I/O on a particular AM-318-10 board, while leaving Super I/O enabled on your other I/O boards. The procedure for doing this is as follows:

1. First, make a copy of your A31810.IDV file; make sure the copy uses a name other than A31810.IDV. For example, while in the DVR: account, type:

   COPY A3181S.IDV=A31810.IDV ENTER

2. Create a copy of your system initialization command file:

   LOG 1,4 ENTER
   COPY TEST.INI=AMOS32.INI ENTER

3. Use AlphaVUE to modify the TEST.INI file. Locate the AM-318-10 board or boards for which you want to disable Super I/O. For all of the TRMDEF statements for the ports on those boards, substitute **A3181S** in place of A31810. Do this only for the AM-318-10 boards for which you want to disable Super I/O.

4. Now, you must add the **/O** switch to the very first TRMDEF statement that uses the AM359S driver. For example:

   `TRMDEF TERM1,A3181S/O=20:19200,AM65,100,100,100`

3. Save the file and leave AlphaVUE. The use the MONTST command to boot with the test initialization file, as described in Chapter 5.

To disable Super I/O for ports on an AM-318-00 or -02 board, use this procedure for AM318.IDV instead of AM31810.IDV.

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